

Household's Problem

Pascal Michailat
<https://pascalmichailat.org/w/>

Household's problem. Choose $c(t)$ & $w(t)$ to maximize

$$\int_0^{\infty} e^{-\delta t} \left[\frac{\varepsilon}{\varepsilon-1} c(t)^{\frac{\varepsilon-1}{\varepsilon}} + \sigma(w(t) - \bar{w}(t)) \right] dt$$

subject to $\dot{w}(t) = r(t)w(t) + a[1-u(t)]h$
 $- [1 + \tau(\theta(t))]c(t) - T(t)/p(t)$

+ no Ponzi condition.

Household takes as given $\theta(t)$, $u(t)$, $p(t)$, $T(t)$, $i(t)$, and initial real wealth $w(0)$.

Hamiltonian (current value).

$$H(t, \overset{\text{control}}{c(t)}, \overset{\text{state variable}}{w(t)}) = \frac{\varepsilon}{\varepsilon-1} c(t)^{\frac{\varepsilon-1}{\varepsilon}} + \sigma(w(t) - \bar{w}(t))$$

$$+ \underset{\text{co-state variable}}{\gamma(t)} [r(t)w(t) + [1-u(t)]ah - [1 + \tau(\theta(t))]c(t) - T(t)/p(t)]$$

Necessary conditions for optimality:

- $\partial H / \partial c = 0$

- $\partial H / \partial w = \delta \gamma(t) - \dot{\gamma}(t)$

- appropriate transversality condition

$$\lim_{t \rightarrow \infty} e^{-\delta t} \gamma(t) w(t) = 0$$

Theorem 7.13 in Acemoglu (2009)

Theorem 7.14 in Acemoglu (2009) $\rightarrow$ any interior solution to necessary conditions is global maximum.

Euler equation:

$$\bullet \partial H / \partial c = 0 \Rightarrow \frac{\varepsilon}{\varepsilon-1} \times \frac{\varepsilon-1}{\varepsilon} \times c^{-1/\varepsilon} - \gamma(t) [1 + \tau(\phi(t))] = 0$$

$$\Rightarrow c(t)^{-1/\varepsilon} = \gamma(t) [1 + \tau(\phi(t))]$$

$$\bullet \partial H / \partial w = \delta \cdot \bar{\gamma}(t) - \dot{\gamma}(t) \Rightarrow \sigma'(w(t) - \bar{w}(t)) + \gamma(t) \tau(t) = \delta \bar{\gamma}(t) - \dot{\gamma}(t)$$

$$\Rightarrow \dot{\gamma}(t) = [\delta - \tau(t)] \bar{\gamma}(t) - \sigma'(w(t) - \bar{w}(t))$$

Special case $w / \tau = 0$, $\sigma' = 0$:

$$\begin{cases} c(t)^{-1/\varepsilon} = \gamma(t) \\ \dot{\gamma} = [\delta - \tau(t)] \gamma(t) \end{cases}$$

Standard Euler equation
 $\swarrow$

$$\begin{cases} \dot{\gamma} / \gamma = \delta - \tau \\ -1/\varepsilon \dot{c} / c = \dot{\gamma} / \gamma \end{cases} \Rightarrow \dot{c} / c = \varepsilon (\tau - \delta)$$